

ANKIT SARKAR

AI Solutions Architect | Platform Engineering | Cloud, DevEx & AI-Assisted Software Delivery

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PROFESSIONAL SUMMARY

AI Solutions Architect and Engineer with 13+ years of experience designing enterprise platforms, cloud-native architectures, and production-grade Generative AI systems on Microsoft Azure. Specialises in agentic AI, multi-agent systems, RAG, MCP, A2A patterns, Azure AI Foundry, Azure OpenAI, Anthropic Claude, and Microsoft Agent Framework. Combines expertise in prompt engineering, LLMops, event-driven knowledge ingestion, vector databases, semantic search, fine-grained authorisation (SpiceDB), observability, model evaluation, and Responsible AI governance. Proven record delivering mission-critical distributed systems, including an Azure eCommerce platform processing 700,000+ orders weekly across 600+ stores. Microsoft Certified Azure Solutions Architect Expert, Certified Agentic AI Business Solutions Architect, and Anthropic Claude Certified Architect.

KEY ACHIEVEMENTS

- Architected production-grade Agentic AI capabilities using Microsoft Agent Framework, Azure AI Foundry, Model Context Protocol (MCP), Agent2Agent (A2A), event-driven knowledge ingestion, Retrieval-Augmented Generation (RAG), vector databases, and SpiceDB to deliver secure, enterprise-scale AI solutions.
- Implemented enterprise AI governance using the Microsoft Agent Governance Toolkit, establishing zero-trust agent identity, policy enforcement, auditable agent actions, and responsible AI controls for production-ready autonomous systems.
- Evaluated and integrated foundation models through Azure AI Foundry, including Azure OpenAI and Anthropic Claude, defining multi-model AI architecture, model selection strategies, and enterprise AI solution patterns based on business, cost, and performance requirements.
- Architected and delivered a Major UK Grocery Retailer's Azure-based eCommerce Picking Platform, supporting 700,000+ customer orders per week across 625 stores, with proven resilience during peak trading events of 90,000+ orders within 30 minutes and zero critical production incidents.
- Led Azure cloud modernisation and platform architecture initiatives across enterprise applications supporting 25,000+ users, establishing reusable architecture patterns, DevSecOps practices, Infrastructure as Code, and engineering standards that improved scalability, resilience, and delivery efficiency.

CORE SKILLS

Agentic AI & Multi-Agent Systems

AI Solution Design • Enterprise AI Platforms • Agentic AI • Multi-Agent Systems • Microsoft Agent Framework • Azure AI Foundry • Azure OpenAI • Anthropic Claude • Model Context Protocol (MCP) • Agent2Agent (A2A) • Claude Agent SDK • Prompt Engineering • Context Engineering • LLMops • Foundation Model Selection • Multi-Model AI Architecture

RAG, AI Security & Governance

Event-Driven Knowledge Ingestion • Vector Databases • Semantic Search • Embedding Models • Reranking • Microsoft Agent Governance Toolkit • Responsible AI • AI Governance • AI Guardrails • AI Security • Fine-Grained Authorization • Relationship-Based Access Control (ReBAC) • SpiceDB • Zero Trust AI • Enterprise AI Compliance

Enterprise & Platform Architecture

Platform Strategy • Solution & System Design • Distributed Systems • Event-Driven Architecture (EDA) • Microservices • Domain-Driven Design (DDD) • API Management & Design • Azure Landing Zones & Governance • Well-Architected Framework • Microsoft Azure • C# • .NET Core / ASP.NET Core • RESTful APIs • Azure Integration Services (APIM, Service Bus, Event Grid, Event Hubs, Azure Functions, Durable Task) • Entity Framework

Platform Engineering, DevSecOps & Observability

Infrastructure as Code (IaC / Terraform) • CI/CD Pipelines (GitHub Actions, Azure DevOps) • Internal Developer Platforms (IDP) • Developer Experience (DevEx) • Containerization (Docker) • InnerSource Practices • Site Reliability Engineering (SRE) • Enterprise Observability (Azure Monitor, App Insights, Log Analytics, OpenTelemetry) • AI Observability • AI Gateway (Azure API Management, Zuplo) • Cloud FinOps & Cost Optimization • High-Availability & Resilience Engineering

Technical Leadership & Governance

Technical Ownership • Architecture Governance • Stakeholder Management • Cross-Functional Leadership • Engineering Standards • Mentoring & Technical Enablement

PROFESSIONAL EXPERIENCE

Tata Consultancy Services

Principal Engineer | Azure Solution Architect

India, Belgium & United Kingdom

March 2019 – Present

Led cloud migrations, application modernization, and platform engineering for global enterprise clients, establishing architectural governance and reusable DevOps standards across cross-functional teams.

Major UK Grocery Retailer – Azure Engineering SME

AI & Azure Solution Architect | Principal Engineer | January 2023 – Present

- Architected secure Agentic AI solutions using Microsoft Agent Framework, Azure AI Foundry, Model Context Protocol (MCP), and Agent2Agent (A2A), designing production-ready patterns for multi-agent interoperability.
- Designed enterprise Retrieval-Augmented Generation (RAG) capabilities spanning event-driven knowledge ingestion, vector search, semantic retrieval, and fine-grained authorization via SpiceDB for secure, context-aware AI experiences.

- Evaluated and prototyped enterprise AI capabilities across Azure AI Foundry, Azure OpenAI, and Anthropic Claude, delivering reusable accelerators and architecture patterns that informed production adoption.
- Established AI governance using the Microsoft Agent Governance Toolkit, implementing policy enforcement, zero-trust agent identities, responsible AI controls, and end-to-end observability for autonomous systems.
- Served as Technical Owner for a major UK retailer's Azure-based eCommerce picking platform supporting 700,000+ weekly orders across 600+ stores, owning platform strategy and event-driven Azure architecture aligned with the Well-Architected Framework.
- Led end-to-end engineering delivery — solution architecture, Terraform-based IaC, DevSecOps, and production operations, improving developer experience through shared Terraform modules and CI/CD workflows, while maintaining reliable operations during peak events including 90,000+ orders in 30 minutes and 150,000+ Christmas orders.

Major UK Health & Beauty Retailer – Azure Cloud Modernisation

Azure Solution Architect | January 2021 – December 2022

- Led Azure architecture and cloud modernisation for 7 business-critical applications supporting over 25,000 internal users, designing Azure-native solutions using Azure App Services, API Management, Key Vault, Azure Storage, and Azure DevOps, with a strong focus on resilience, security, and cost-aware architecture.
- Designed and delivered a reusable Stub Identity Platform that removed a critical programme dependency and enabled enterprise-scale performance testing without impacting production identity services, alongside reusable frameworks, shared platform services, and engineering accelerators adopted across multiple delivery teams.
- Partnered with Microsoft architects and client engineering leadership to establish cloud governance and migration strategy across a programme modernising approximately 90 enterprise applications.

Belgian National Railways – Azure Cloud Migration

Azure Migration Architect | March 2019 – December 2020

- Supported a large-scale Azure migration programme for Belgium's national railway operator, covering complex enterprise applications and infrastructure, and defined migration strategies, Azure Landing Zone architecture, and cloud adoption roadmaps aligned with business objectives.
- Contributed to governance, workload discovery, dependency analysis, and migration planning across multiple delivery teams.

MOL Information Technology

Software Engineer

India

January 2016 – February 2019

- Contributed to the development of the MOL Consolidation System (MCS), a business-critical logistics and freight management platform supporting global shipping operations, designing reusable frameworks, NuGet packages, and shared libraries adopted across multiple engineering teams.
- Established engineering standards, coding practices, and application templates to improve consistency and maintainability, and introduced CI/CD pipelines and automated build processes that improved release reliability and reduced manual deployment effort.
- Conducted code reviews, mentored junior developers, and contributed to performance optimisation and modernisation initiatives across legacy .NET applications.

Early Career

India

September 2013 – December 2015

Early career developer roles focused on engineering, supporting, and delivering web applications built on the Microsoft stack within collaborative Agile teams.

Software Developer – Customer First Infonet Pvt Ltd (2014 – 2015)

Software Developer – Max Mobility Pvt Ltd (2013 – 2014)

- Developed enterprise web applications using ASP.NET MVC, C#, SQL Server, and the Microsoft technology stack, delivering new features, production support, and defect fixes across customer-facing applications in Agile teams, while collaborating with business stakeholders to implement enhancements and ensure application stability.
- Gained foundational experience in enterprise software engineering, SDLC processes, and Agile delivery practices.

CERTIFICATIONS

Amazon (www.credly.com/users/sarkaran/badges/credly)

AWS Certified Cloud Practitioner

Anthropic (www.credly.com/users/sarkaran/badges/credly)

Claude Certified Architect - Professional • Claude Certified Architect - Foundations • Claude Certified Architect - Foundations • Claude Certified Associate – Foundations

GitHub (www.credly.com/users/sarkaran/badges/credly)

GitHub Advanced Security • GitHub Actions • GitHub Copilot • GitHub Administration • GitHub Foundations

Microsoft (learn.microsoft.com/en-us/users/ankitsarkar/transcript/vn0qmu4wl2xgwye)

Azure Solutions Architect Expert (AZ-303+AZ-304) • Agentic AI Business Solutions Architect (AI-102 + AB-100) • Azure DevOps Engineer Expert (AZ-400) • Azure Developer Associate (AZ-204) • Azure Administrator Associate (AZ-104) • Azure AI Fundamentals (AI-900) • Azure Fundamentals (AZ-900)

EDUCATION

Bachelor of Technology (B.Tech) in Electronics & Communication Engineering

2009 – 2013

Camellia Institute of Technology, India